

# Anand Sagar

Researcher – Materials Physics & Spectroscopy

Indian Institute of Technology Ropar, India

anand.25phz2025@iitrpr.ac.in | +91-7009639618

## Research Objective

---

Materials physics researcher specialising in thin-film deposition, ion irradiation, and spectroscopic characterisation of nanostructured materials. Seeking to advance functional nanomaterial development for energy storage and sensing applications, integrating data-driven analysis and machine-learning methods into experimental workflows.

## Education

---

**M.Sc. Physics** (Gold Medalist)

2022 – 2024

Chandigarh University, Mohali, India

- CGPA: 8.48 / 10
- Master's thesis: *Structural, Optical, and Photocatalytic Properties of Ni-Zn and Mg-Zn Spinel Ferrites*

**B.Sc. Physics**

2019 – 2022

Panjab University, Chandigarh, India

- Percentage: 79.9%

## Research Experience

---

**Research Intern – Functional Materials and Spectroscopy**

2025 – Present

Indian Institute of Technology Ropar, India

- Thin-film deposition using DC/RF magnetron sputtering.
- Ion irradiation experiments using KDC-40 ion source.
- Fabrication and characterisation of nanostructured materials for energy and sensing applications.
- Multivariate analysis of spectroscopic datasets (Raman, FTIR, UV-Vis).
- Application of Principal Component Analysis (PCA) for dimensionality reduction and spectral feature extraction.
- Supervised ML models – Convolutional Neural Networks (CNN) for spectral pattern recognition; familiarity with neural-network mathematics and Python-based training pipelines.

**Project Fellow – IUAC Funded Project (UFR-75311)**

Oct 2024 – Jul 2025

Inter-University Accelerator Centre (IUAC), New Delhi

- Fabrication of cobalt-doped MoS<sub>2</sub> thin films on stainless-steel substrates.
- Ag<sup>12+</sup> ion irradiation using Pelletron accelerator.
- Structural and electrochemical analysis of irradiated materials for supercapacitor electrode applications.
- Correlation of irradiation-induced structural modifications with optical and electronic properties.

**Project Fellow – Hydrogen Evolution Reaction (HER)**

Mar 2024 – Jul 2024

GLA University, Mathura (Advisor: Dr. Aayush Gupta, Mech. Engg.)

- Synthesised BST–Graphene Oxide composite materials for catalytic hydrogen evolution reactions.
- Investigated electrochemical performance, catalytic activity, and structure–property correlations.

## Master’s Thesis Research

2023 – 2024

Chandigarh University (Supervisor: Dr. Sumit Bhardwaj)

- Synthesis and characterisation of Ni–Zn and Mg–Zn spinel ferrites.
- Structural analysis via XRD (FullProf Rietveld refinement); optical characterisation via UV–Vis and Raman spectroscopy.
- Investigation of photocatalytic properties and bandgap engineering in complex spinel ferrites.

## Research Interests

- 
- Functional nanomaterials for spectroscopy and sensing
  - Light–matter interaction and plasmonics
  - Spectroscopic sensing technologies
  - Machine learning in spectroscopy and materials science
  - Multiferroic and magnetodielectric materials
  - Electrochemical energy storage (supercapacitors, HER electrocatalysis)

## Technical Skills

---

|                              |                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Thin-Film Fabrication</b> | DC/RF magnetron sputtering, spin coating                                                         |
| <b>Characterisation</b>      | XRD, SEM, TEM, AFM, XPS, Raman spectroscopy, FTIR, UV–Vis spectroscopy                           |
| <b>Programming</b>           | Python (NumPy, SciPy, scikit-learn, TensorFlow/Keras), PCA and multivariate statistical analysis |
| <b>Scientific Software</b>   | OriginPro, FullProf (Rietveld refinement), HighScore Plus, VESTA                                 |
| <b>Techniques</b>            | Pelletron accelerator operation, ion irradiation, electrochemical measurements (CV, EIS)         |

## Publications

- 
1. **A. Sagar**, A. Bansal, P. Dubey, K. Yadav, S. Singh, A. Kumar, P. Shahi, R. Chandra, and R. Walia, “Engineering cobalt-doped MoS<sub>2</sub> nanostructures on stainless steel for cost-effective supercapacitor electrodes,” *Ionics*, 2026. DOI: [10.1007/s11581-026-XXXXX](https://doi.org/10.1007/s11581-026-XXXXX) [Published 11 May 2026]
  2. **A. Sagar**, S. Bhardwaj, A. Gupta, H. Tripathi, and R. Shukla, “Structural, optical, and photocatalytic properties of Ni–Zn and Mg–Zn spinel ferrites,” *Hyperfine Interactions*, **245**, 86 (2024). [First Author]  
DOI: [10.1007/s10751-024-01930-0](https://doi.org/10.1007/s10751-024-01930-0)
  3. S. Rani, **A. Sagar**, A. R. Khan, A. Gupta, and S. Bhardwaj, “Investigations on structural, dielectric, and ferroelectric properties of Bi<sub>4-x</sub>Nd<sub>x</sub>Ti<sub>3</sub>O<sub>12</sub>,” *Hyperfine Interactions*, **245**, 251 (2024).  
DOI: [10.1007/s10751-024-02082-x](https://doi.org/10.1007/s10751-024-02082-x)
  4. S. Rani, A. R. Khan, **A. Sagar** et al., “Structural, dielectric and magnetodielectric behavior of Bi<sub>3.15</sub>Nd<sub>0.85</sub>Ti<sub>3-x</sub>(Fe<sub>x/2</sub>Cr<sub>x/2</sub>)O<sub>12</sub> ceramics,” *Ceramics International*, **51**, 8558–8569 (2025).  
DOI: [10.1016/j.ceramint.2024.12.286](https://doi.org/10.1016/j.ceramint.2024.12.286)
  5. S. Rani, A. R. Khan, **A. Sagar** et al., “Tuning magnetodielectric coupling in Bi<sub>3.15</sub>Nd<sub>0.85</sub>Ti<sub>3-x</sub>(Fe<sub>x/2</sub>Cr<sub>x/2</sub>)O<sub>12</sub>

ceramics,” *Ceramics International*, 2026.

DOI: [10.1016/j.ceramint.2026.02.139](https://doi.org/10.1016/j.ceramint.2026.02.139)

## Conference Presentations

---

- Oral/Poster presentation at the *International Conference on Materials Science and Nanotechnology*. Presented research on multiferroic ceramics and magnetodielectric materials.
- Presentation at an international conference on spectroscopy and functional materials.

## Awards and Achievements

---

- **Gold Medalist** – M.Sc. Physics, Chandigarh University (2024)
- 2 First-author peer-reviewed publications
- Five peer-reviewed international journal publications within two years of research career
- Access to national facility – IUAC Pelletron accelerator (competitive grant UFR-75311)

## References

---

**Dr. Subhendu Sarkar**

Associate Professor

IIT Ropar

[sarkar@iitrpr.ac.in](mailto:sarkar@iitrpr.ac.in)

**Dr. Sumit Bhardwaj**

Associate Professor

Chandigarh University

[sumit.bhardwaj4@gmail.com](mailto:sumit.bhardwaj4@gmail.com)

**Dr. Rajan Walia**

Assistant Professor

University of Allahabad

[rwalia@allduniv.ac.in](mailto:rwalia@allduniv.ac.in)